



ProbFuse:

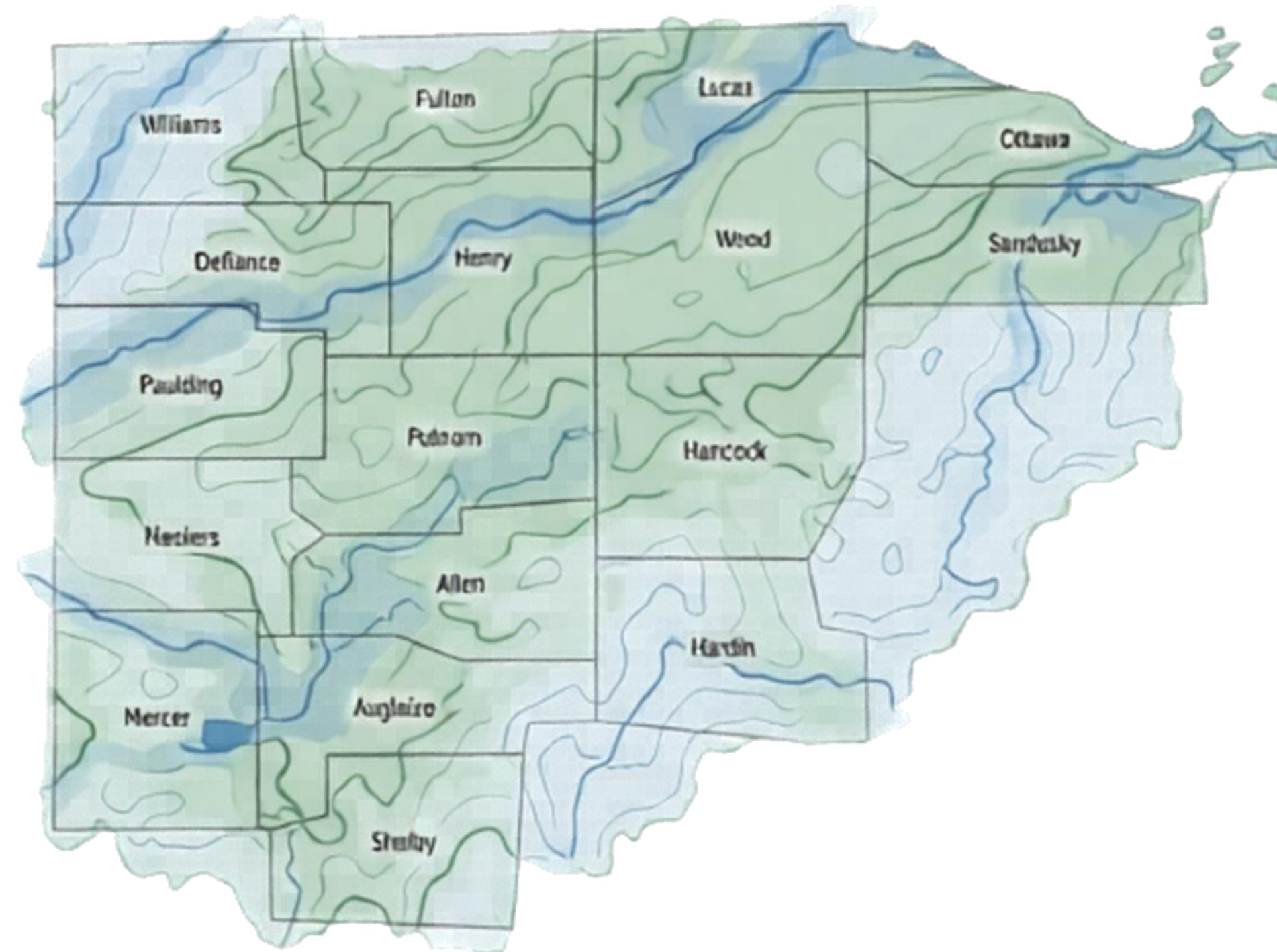
Data-Driven Analysis of Agricultural Risk and Funding Impact using Multi-Kernel Gaussian Process

Linking conservation funding, environmental factors and yield uncertainty for policy-relevant insights in the Ohio Maumee River Basin

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Committee: Dr. Scott Shearer, Dr. Rajiv Ramnath

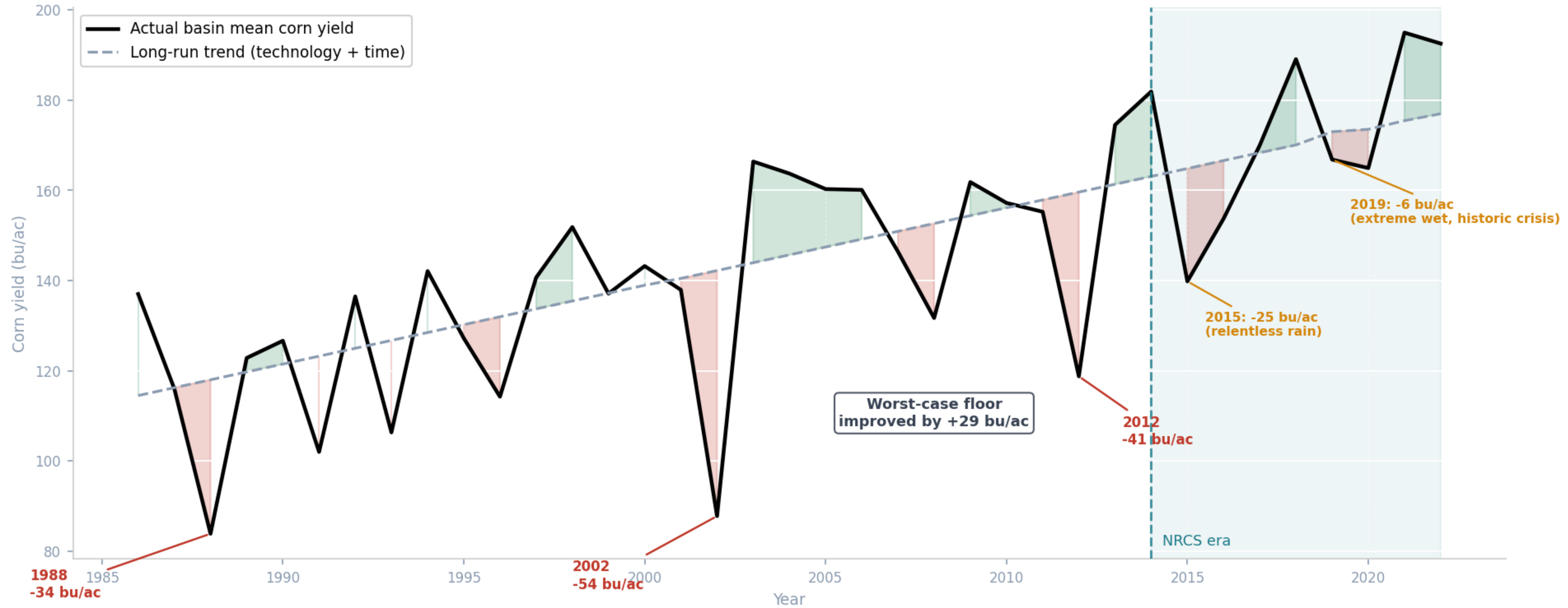
Study Focus: 16 counties in Ohio Maumee River Basin



Why?

- High-priority agricultural watershed → important for policy impact
- Significant conservation funding → allows analysis of alignment with needs
- Varied conditions across counties → helps test if funding targets vulnerable areas effectively

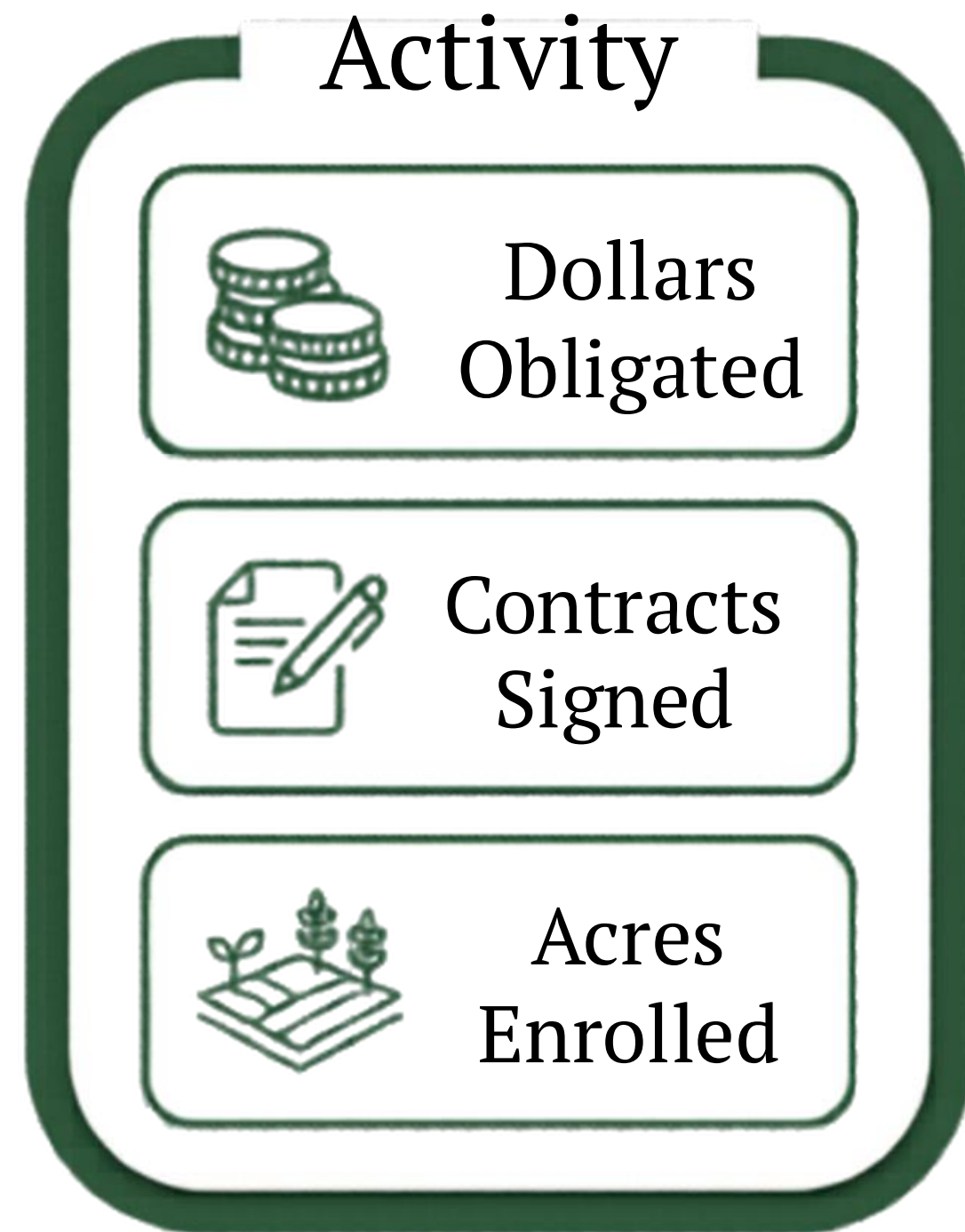
Maumee Basin - 37 Years of Corn Yield: The Bad Years Got Less Bad



Outline

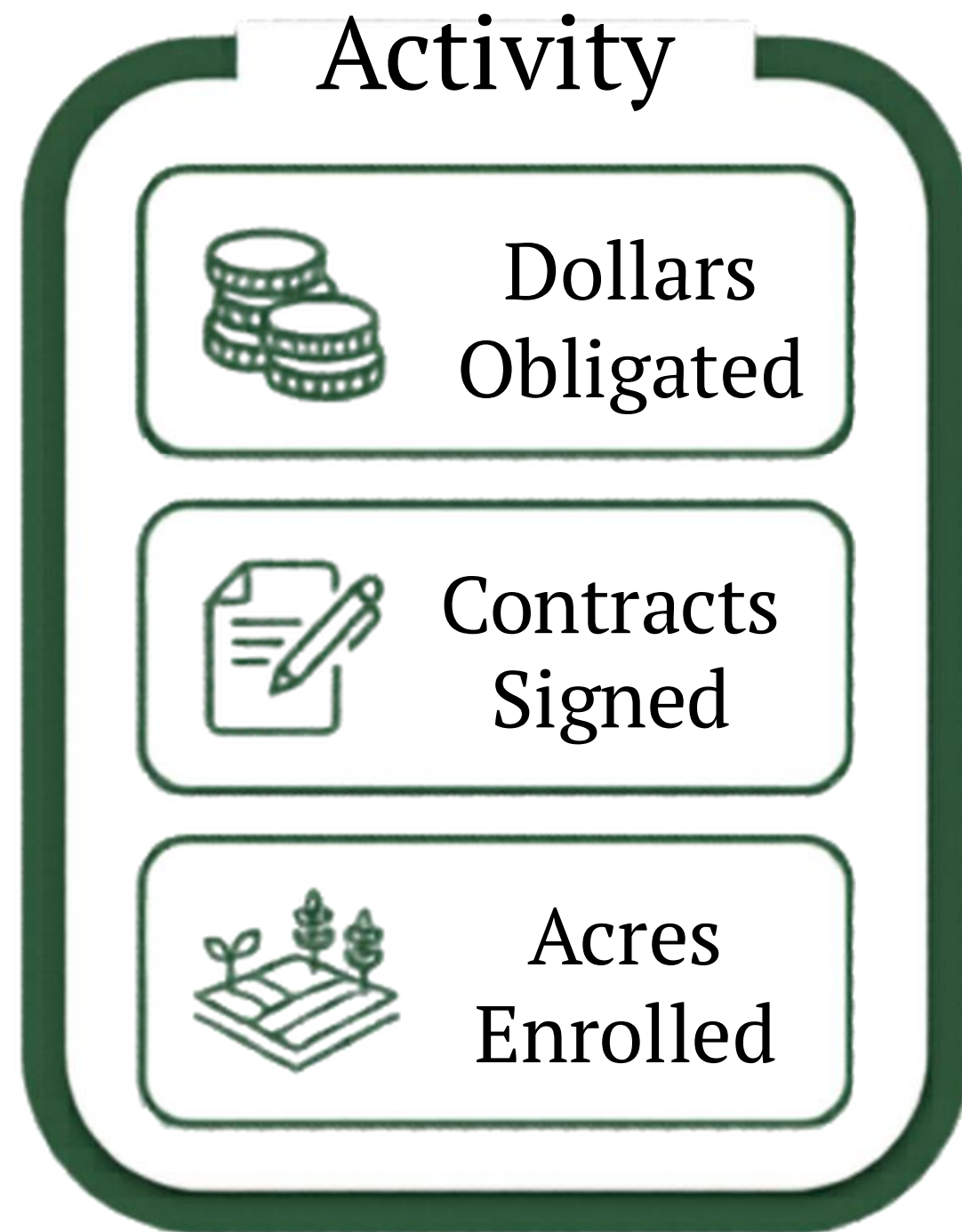
- **Problem Statement**
- Literature Review & Gaps
- Methodology
- Evaluation
- Results
- Contribution
- Who can use?
- Limitation & Future Work
- Conclusion

Problem Statement

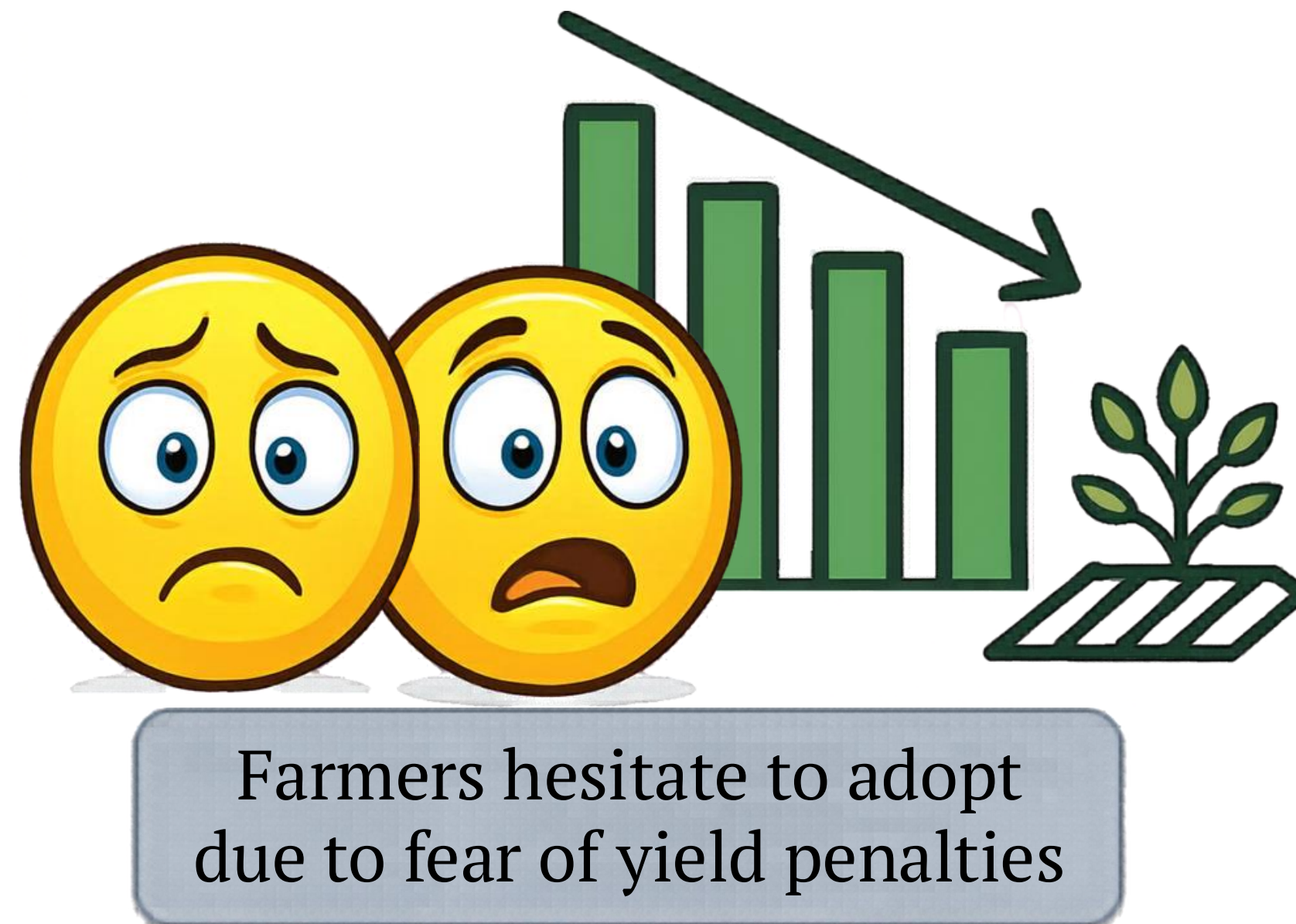


Billions Invested
(EQIP & CSP)

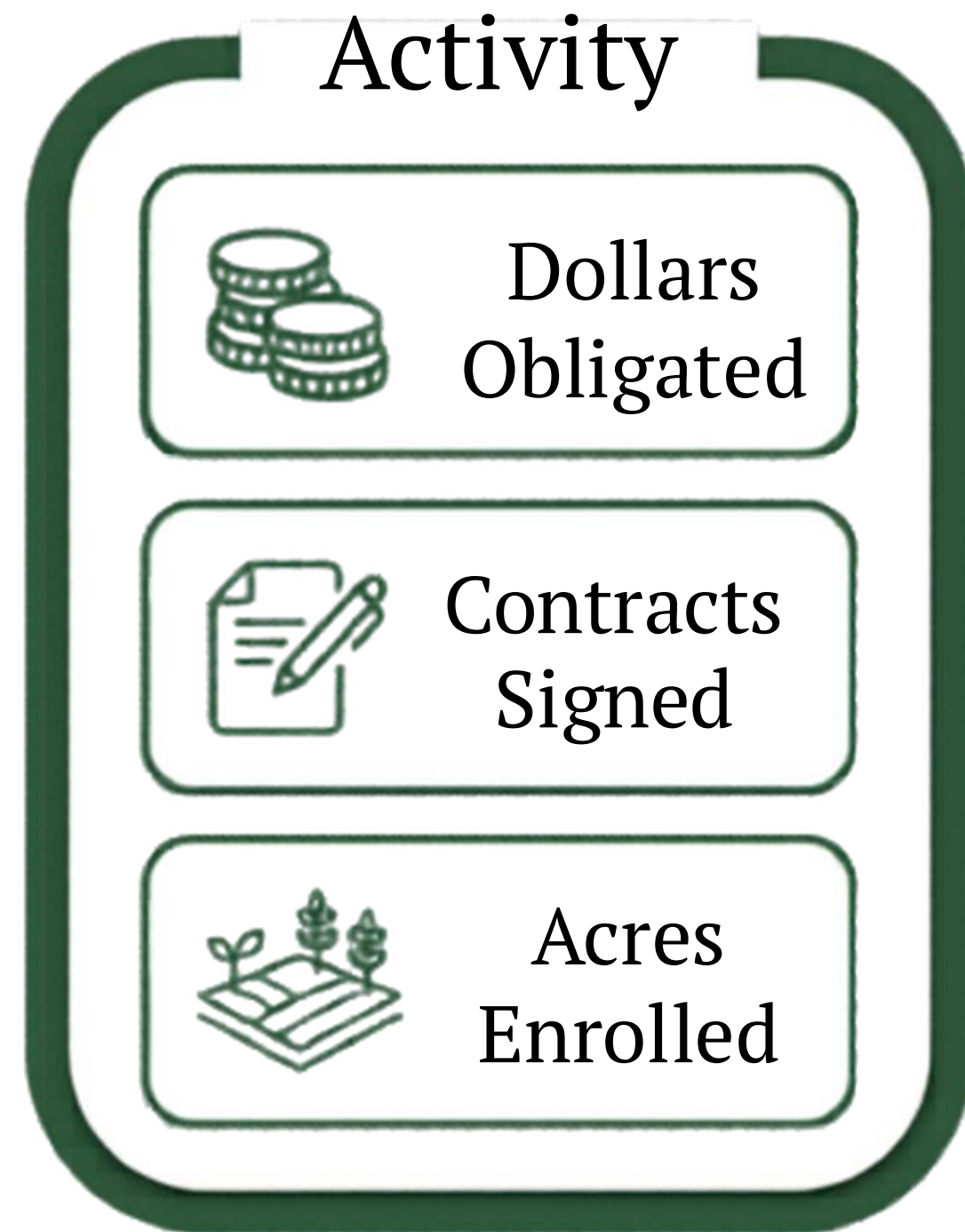
Problem Statement



Billions Invested
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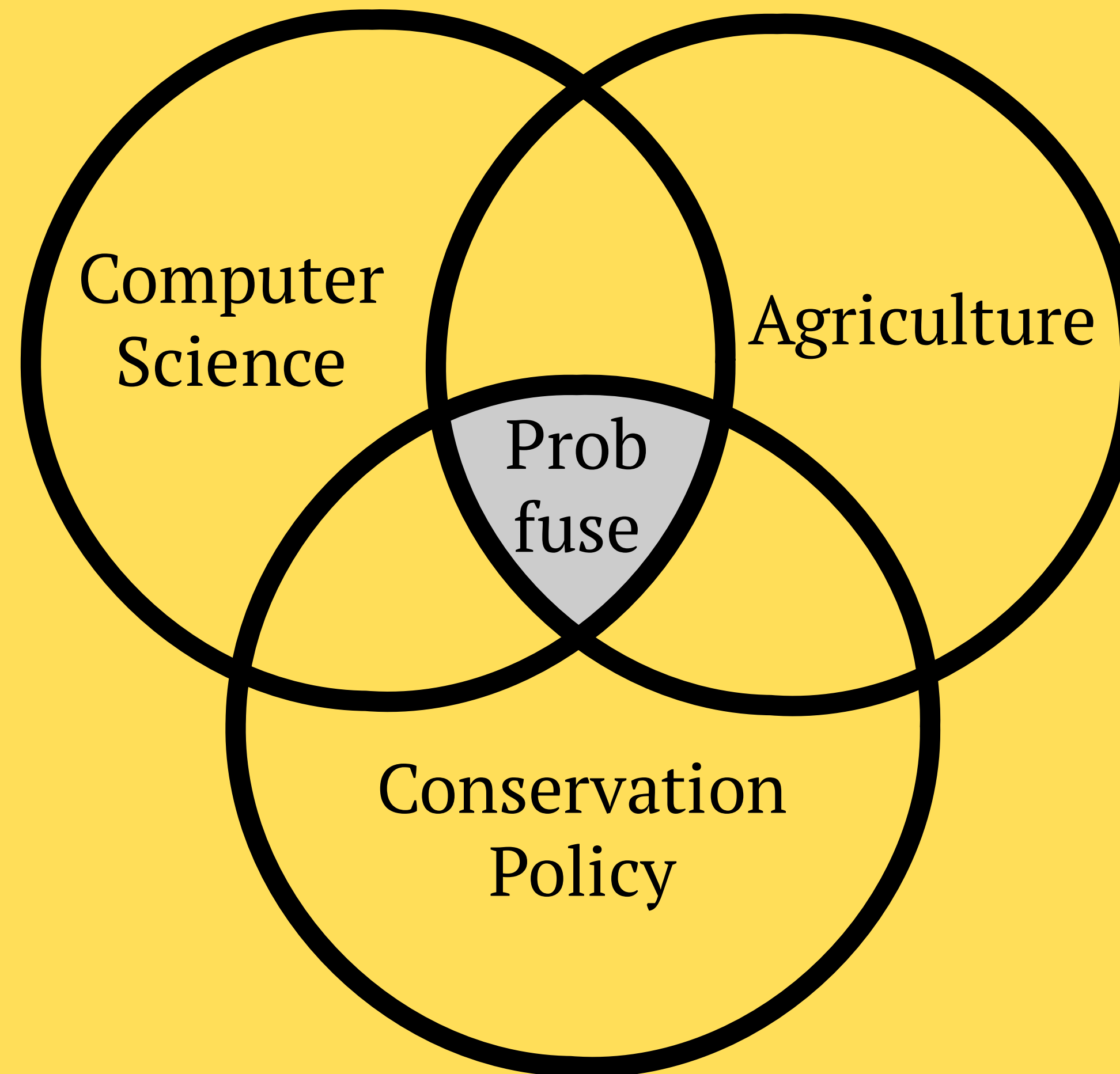


Problem Statement



Billions Invested
(EQIP & CSP)





Core Question: Is this resilience coming from the money that NRCS is spending and is that money reaching the countries that need it most?

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Overcoming Existing Constraints in Agricultural AI



Gap: Small-Data Regime at county-level



ProbFuse: Using Gaussian Processes



Gap: Point Predictions lack confidence



ProbFuse: Produces Calibrated Uncertainty



Gap: Black-Box AI prevents public trust



ProbFuse: Interpretable Multi-Kernel Architecture



Gap: Fragmented Weather & Soil Data



ProbFuse: Unified Public Data Fusion Pipeline



Gap: Static Evaluation of programs



ProbFuse: Delay-Aware Policy Indexing (3-5 year lags)



Gap: Translation Gap from models to action

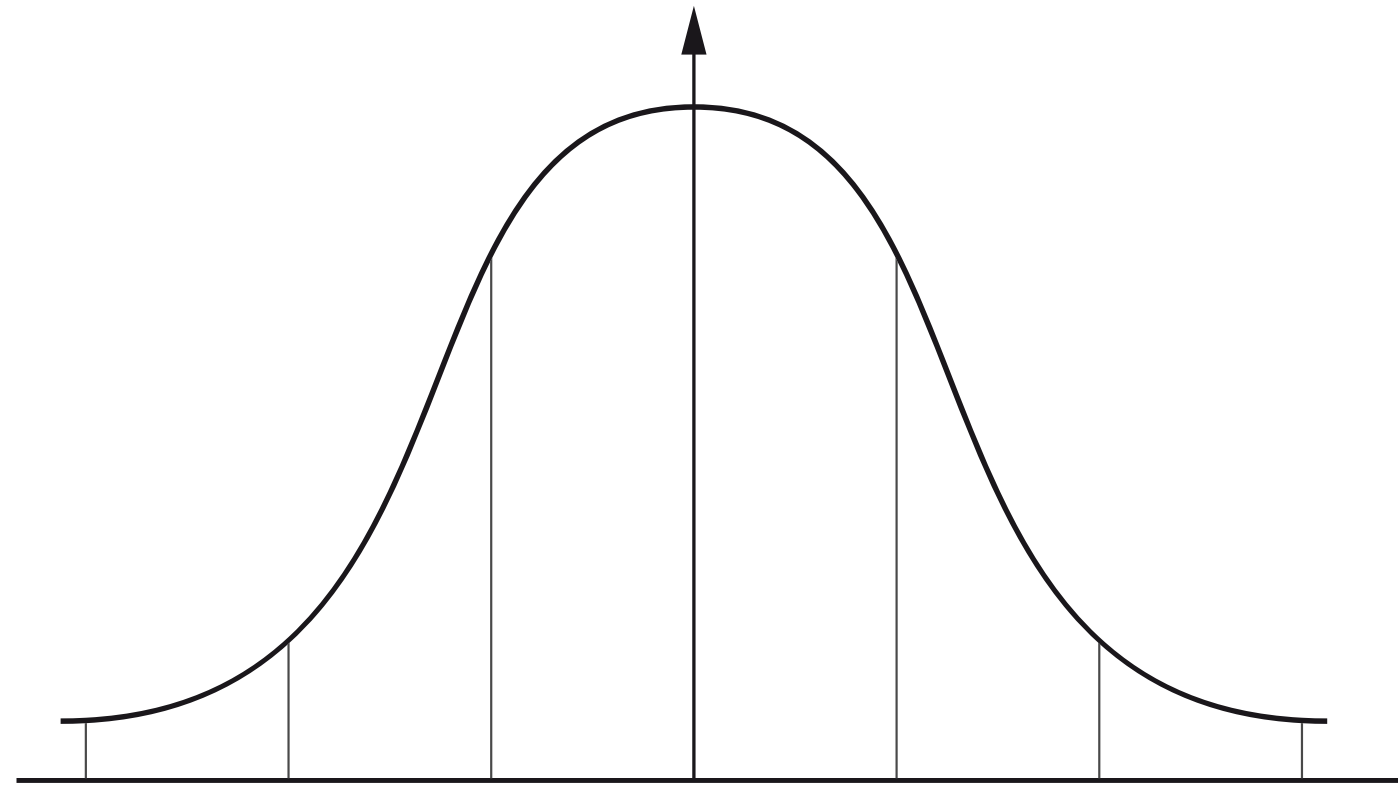


ProbFuse: Actionable 2x2 Quadrant Typology

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ProbFuse: Objective Functions



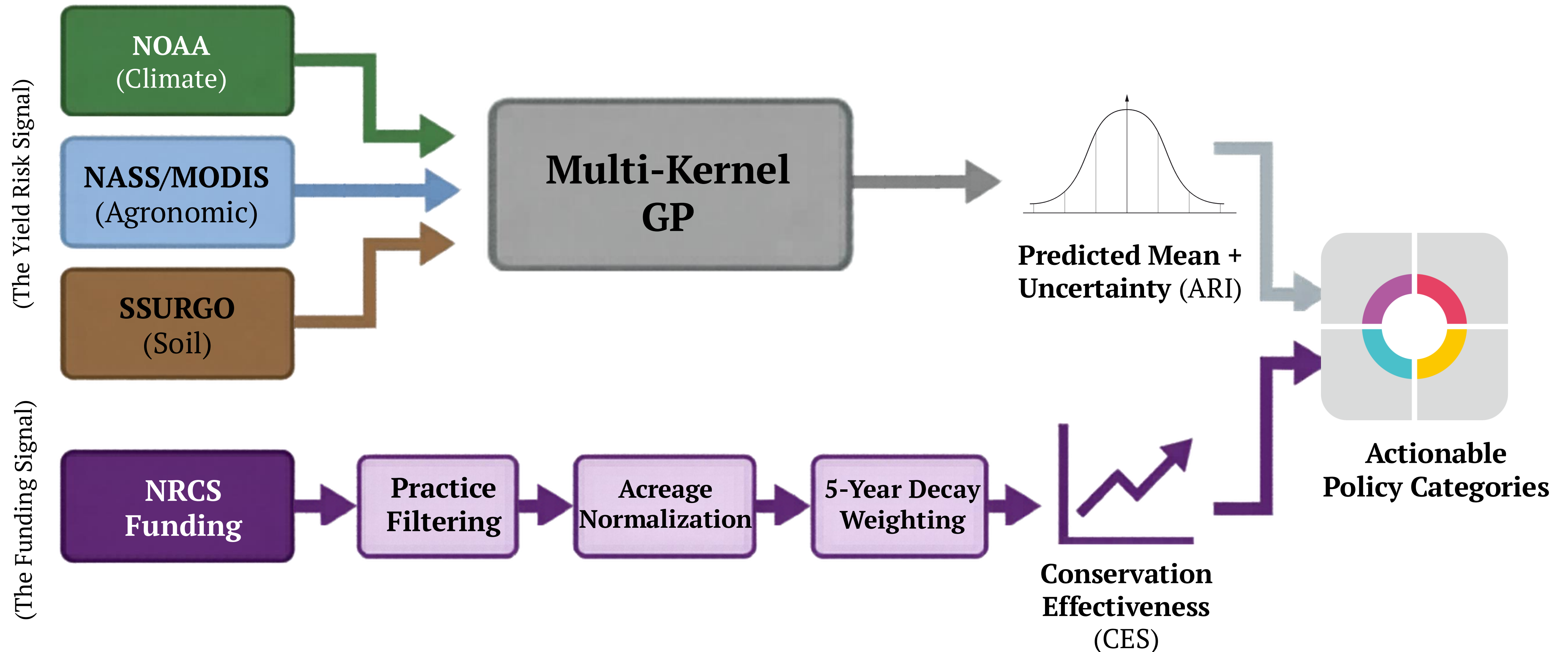
**Yield
Vulnerability
(ARI)**



**Investment
Accumulation
(CES)**

Linking predicted county-level vulnerability with historical conservation accumulation.

The ProbFuse dual-stream architecture



Linking predicted county-level vulnerability with historical conservation accumulation.

Harmonizing homogeneous public data across 36 years

Stage 1: Input Layer

NOAA NCEI
(Station Daily)

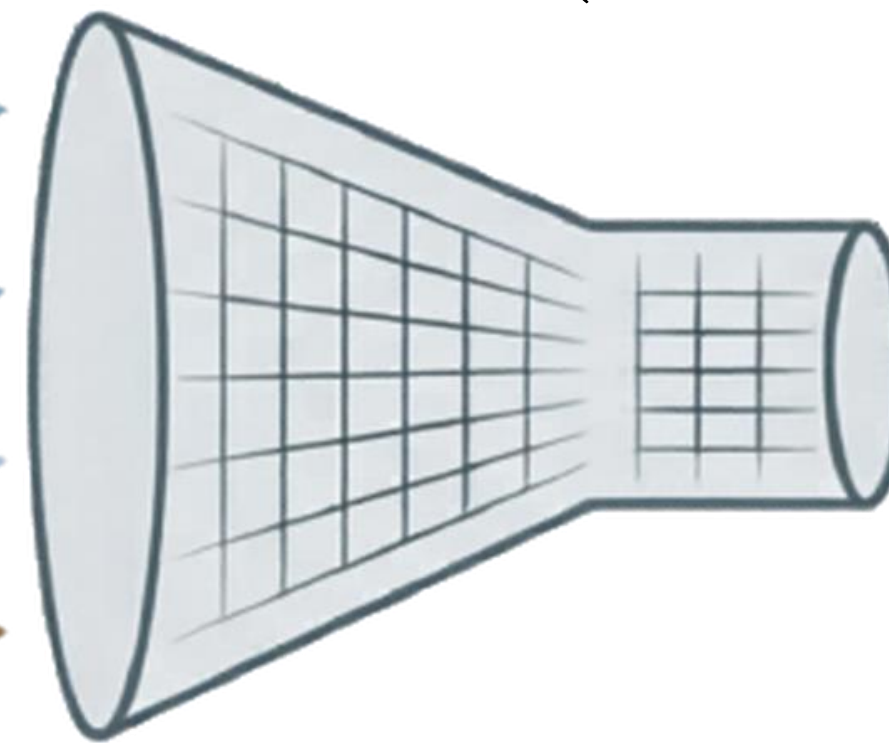
USDA NASS
(County Annual)

MODIS/LANDSAT
(Off-season NDVI)

USDA SSURGO
(Static Map Units)

Stage 2: Alignment Layer

Unified County-Year-Crop
Feature Matrix (1986-2022)



$$Y_{anomaly} = Y_{raw} - Y_{trend}$$

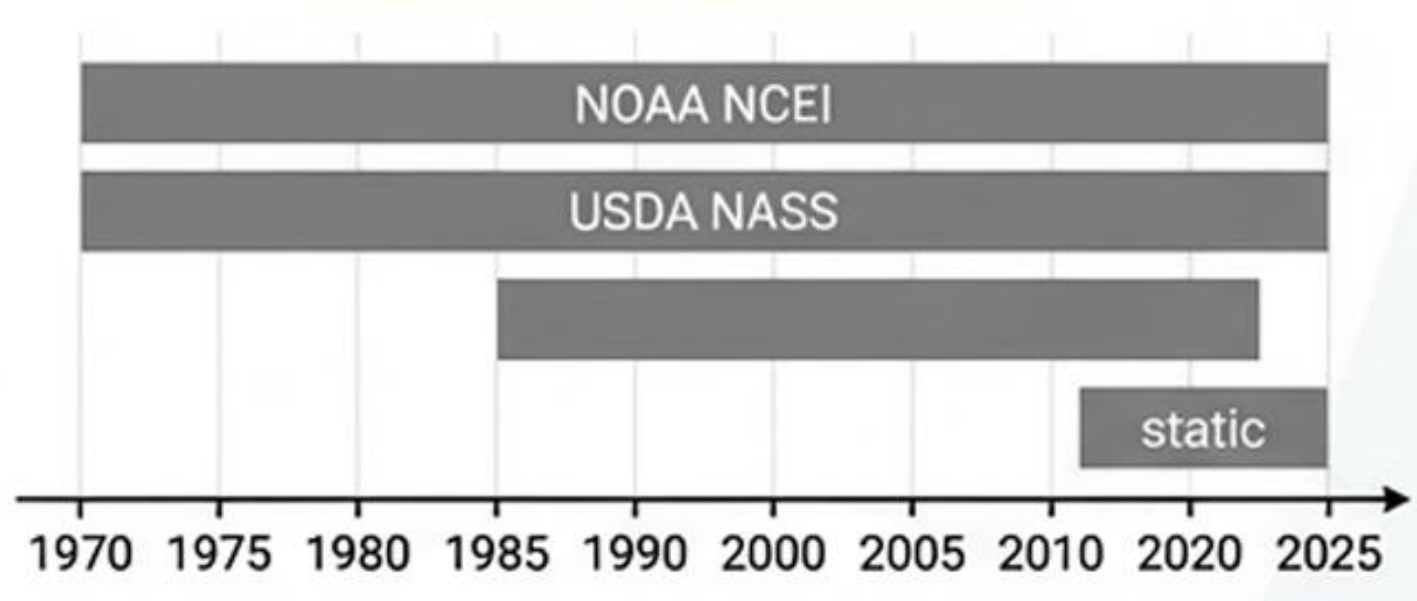
(Detrending isolates weather, soil, and
management from technological gains)

Stage 3: Domain Partition Layer

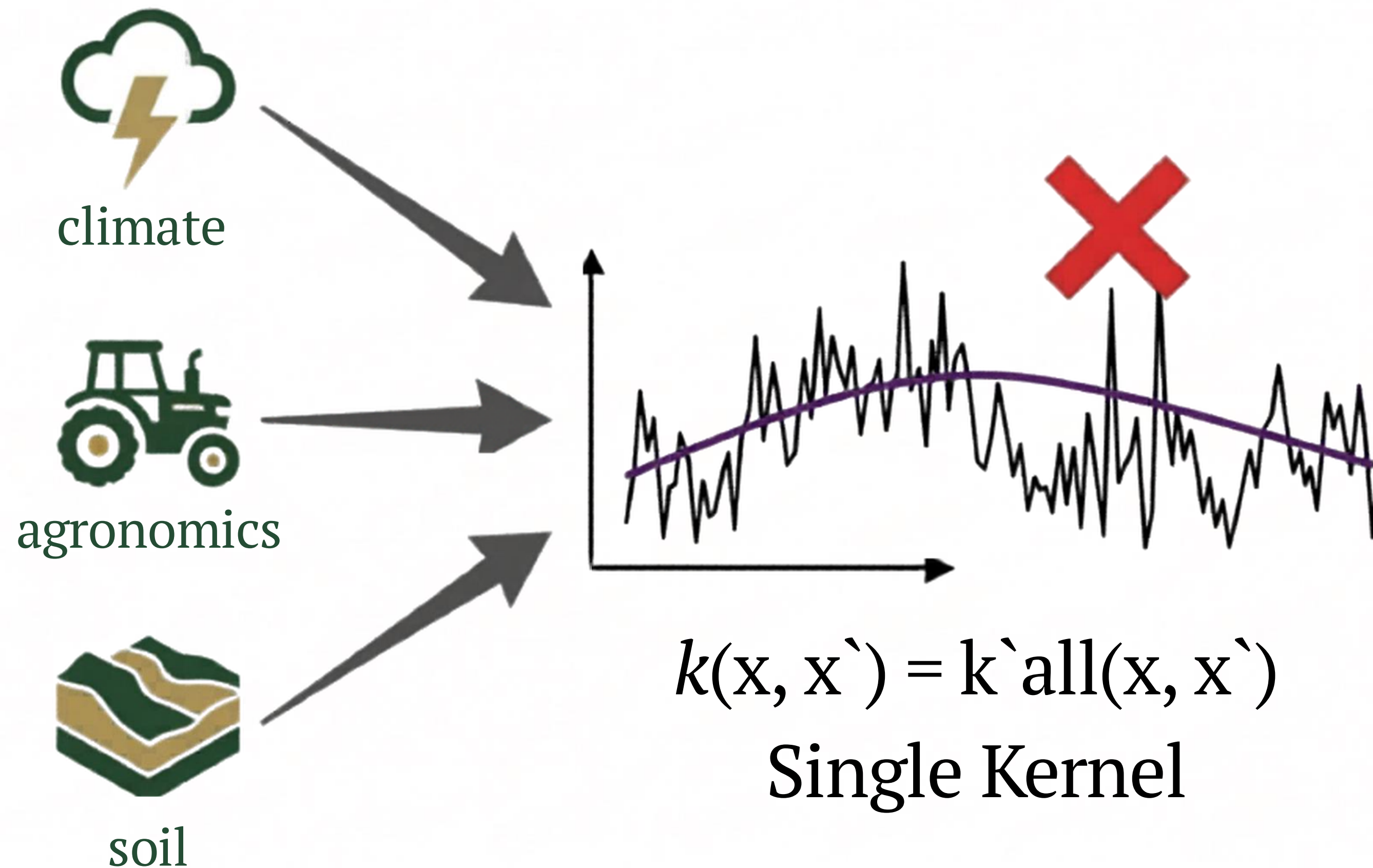
$x \in R_6$
(SPEI, GDD, Precipitation)

$x \in R$
(NDVI Cover Intensity,
Lagged Anomaly, Crop Type)

$x \in R_6$
(AWC, Drainage Class,
Spatial Coords)



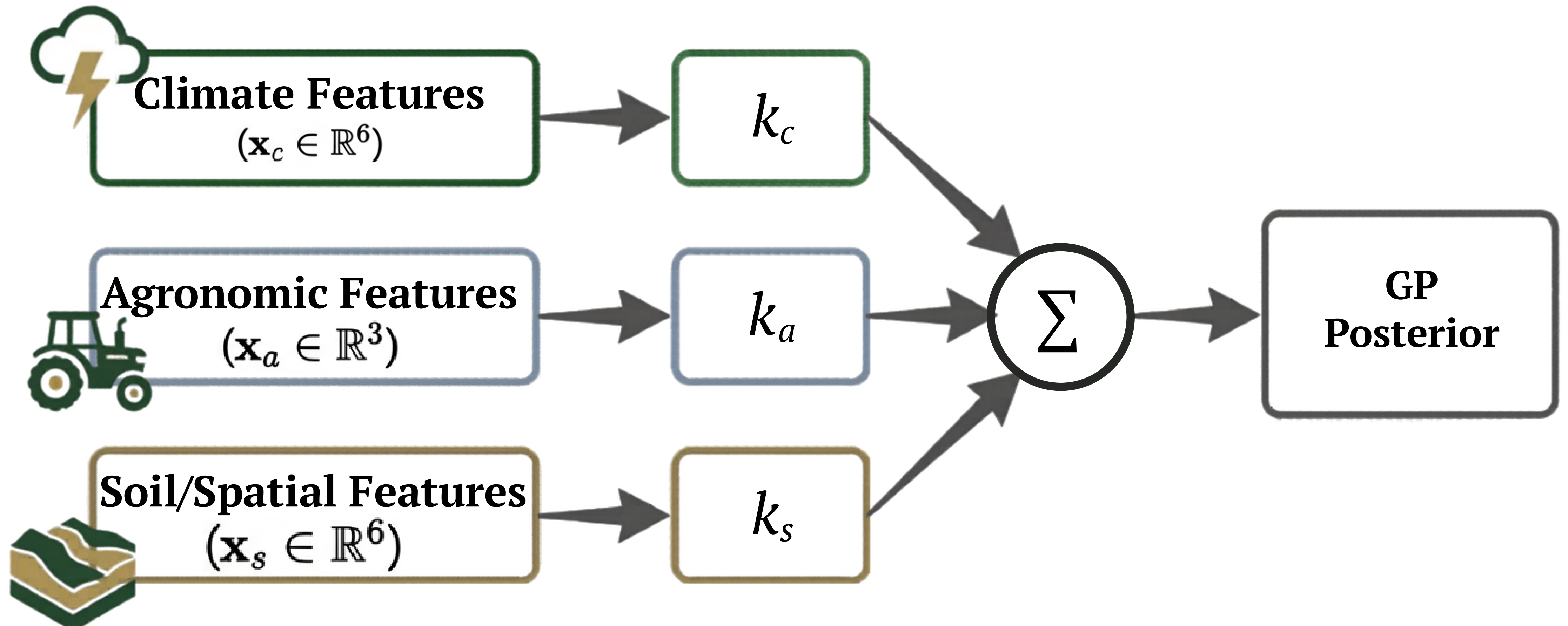
The Structural Flaw in a Single-Kernel Gaussian Process



The Flaw: While a standard GP provides probabilistic uncertainty, a single kernel assumes same smoothness across all features and lacks interpretability

Climate features vary abruptly. Soil features vary smoothly across geography. A principled model must reflect these distinct structural assumptions

Designing a Domain-Decomposed Multi-Kernel



- We explicitly partition the feature space to reflect physical reality
- Preserves domain-separated uncertainty and structural interpretability

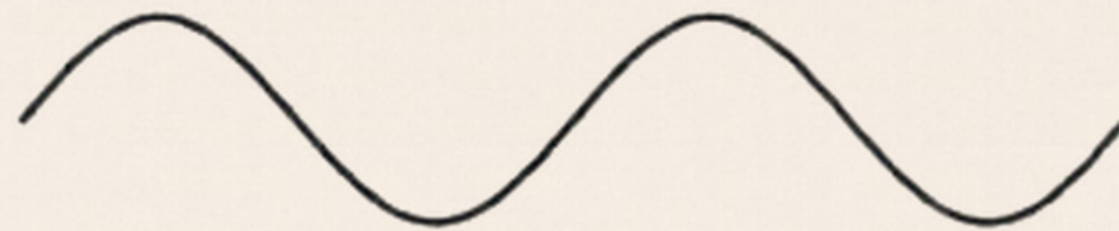
Kernel Structure Selection

$$\text{NLML} = \frac{1}{2}y^T K_y^{-1}y + \frac{1}{2}\log|K_y| + \frac{n}{2}\log(2\pi)$$

Selection Criterion: Negative Log Marginal Likelihood (NLML)

Soil/Spatial

Winning Kernel:
RBF Kernel



Selected for slow,
geographical transitions.

Climate

Winning Kernel:
Matérn-5/2 Kernel



Selected for structured but
abrupt weather anomalies.

Agronomic

Winning Kernel:
Matérn-3/2 Kernel



Selected for discrete crop
states and management lags.

The Final Additive Kernel & Data-Driven Initialization

$$k(\mathbf{x}, \mathbf{x}') = k_c(\mathbf{x}_c, \mathbf{x}'_c) + k_a(\mathbf{x}_a, \mathbf{x}'_a) + k_s(\mathbf{x}_s, \mathbf{x}'_s) + \sigma_n^2 \delta_{ij}$$

L-BFGS-B Initialization:

- Lengthscales: Initialized using median pairwise Euclidean distance
- Signal Variance: Empirical target variance (~246.7)
- Noise: Initialized at 10% of signal

The Additive Advantage:

- The resulting additive covariance allows us to isolate how much variance is driven by climate stress versus soil properties.

The Final Learned Additive Kernel

$$k = 2.27^2 \cdot k_{\text{Mat}5/2}(\ell=10) + 1.11^2 \cdot k_{\text{Mat}3/2}(\ell=10) \\ + 1.87^2 \cdot k_{\text{RBF}}(\ell=5.31) \quad + k_{\text{White}}(\sigma_n^2=0.558)$$

Dominant signal: Weather variability

Marginal signal: Agronomic carry-over

Secondary signal: Persistent soil/spatial

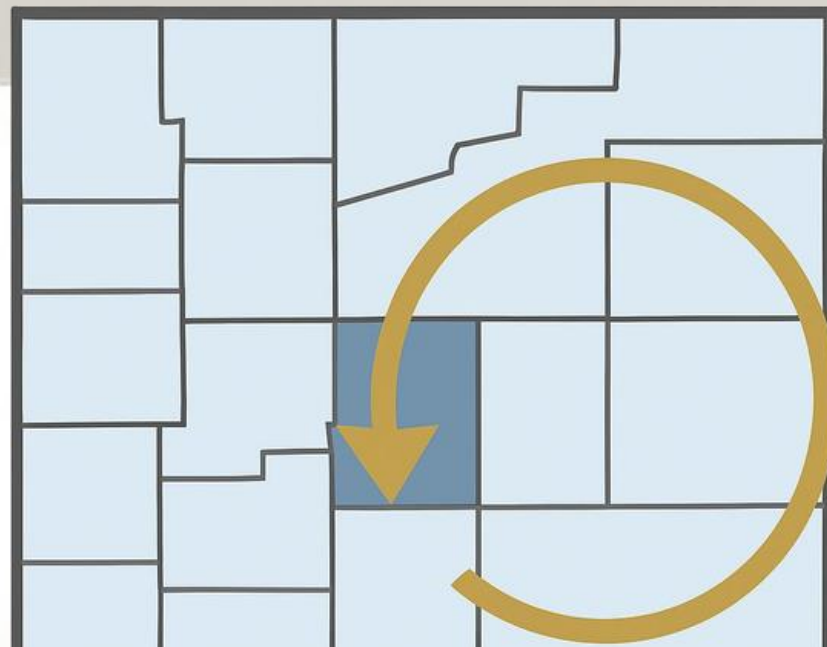
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Generalization: Leave-One-County-Out (LOCO)

LOCO directly simulates deployment to a new geography unseen during training

Metric	Value
CRPS	6.95 bu/ac
Coverage@90	91.9%
RMSE	12.56 bu/ac



Continuous Ranked Probability Score (CRPS)

Evaluates the sharpness and accuracy of the full predictive distribution, not just the mean.

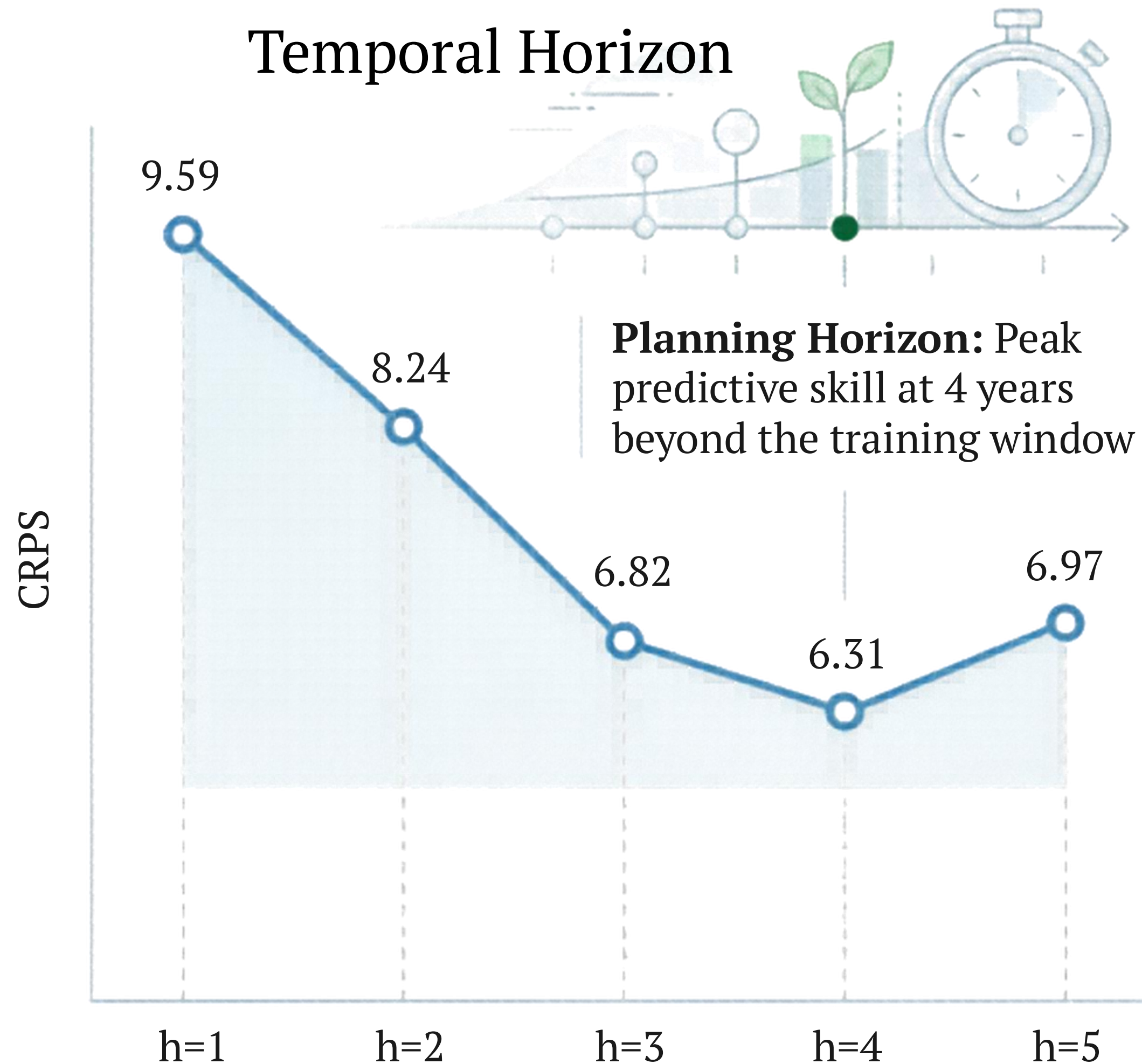
Coverage@90 Calibration

Over many predictions, how often did the actual value actually fall inside your interval?

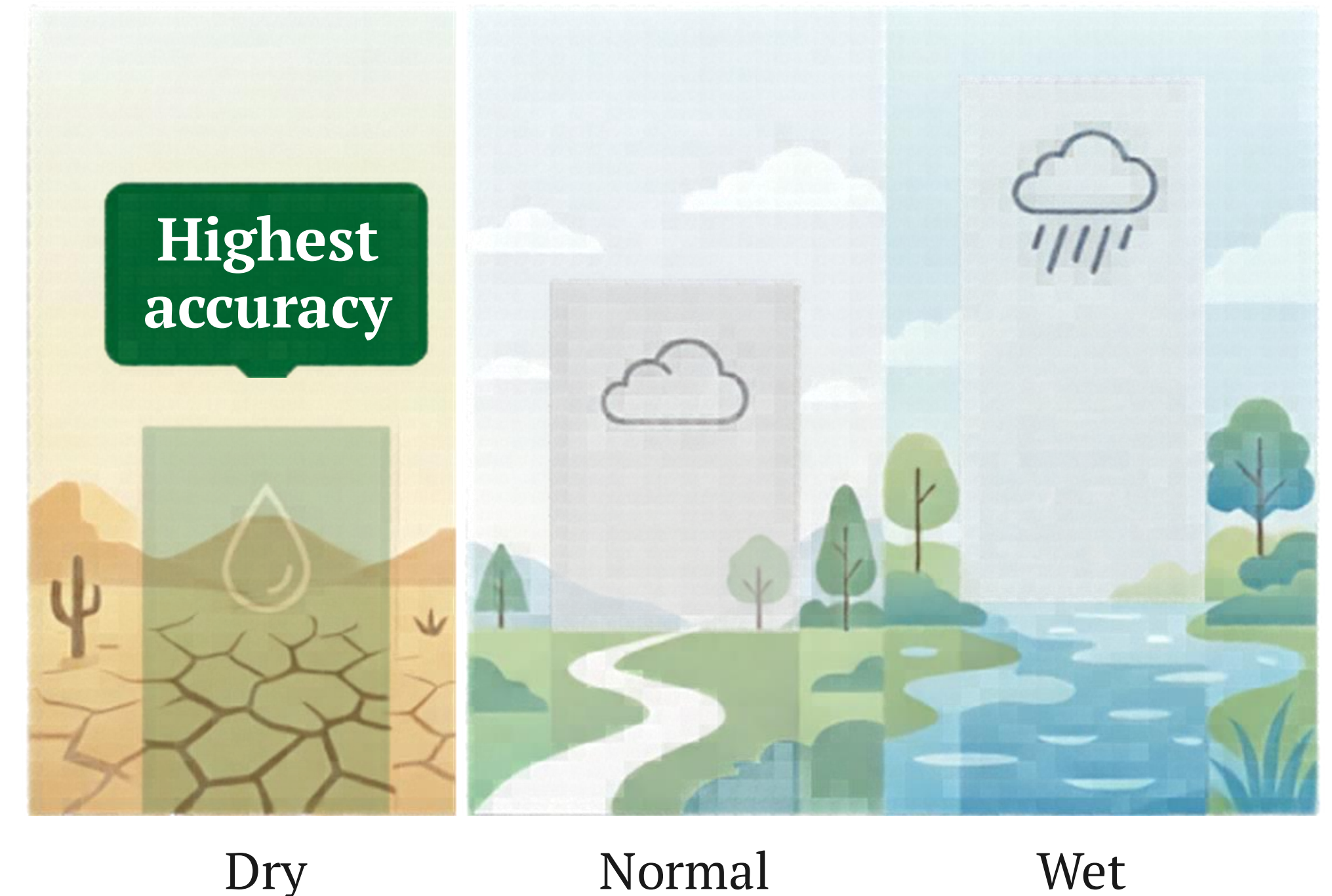
Ground Truth: NASS Yield Data

Temporal Stability and Climate Extremes

Temporal Horizon

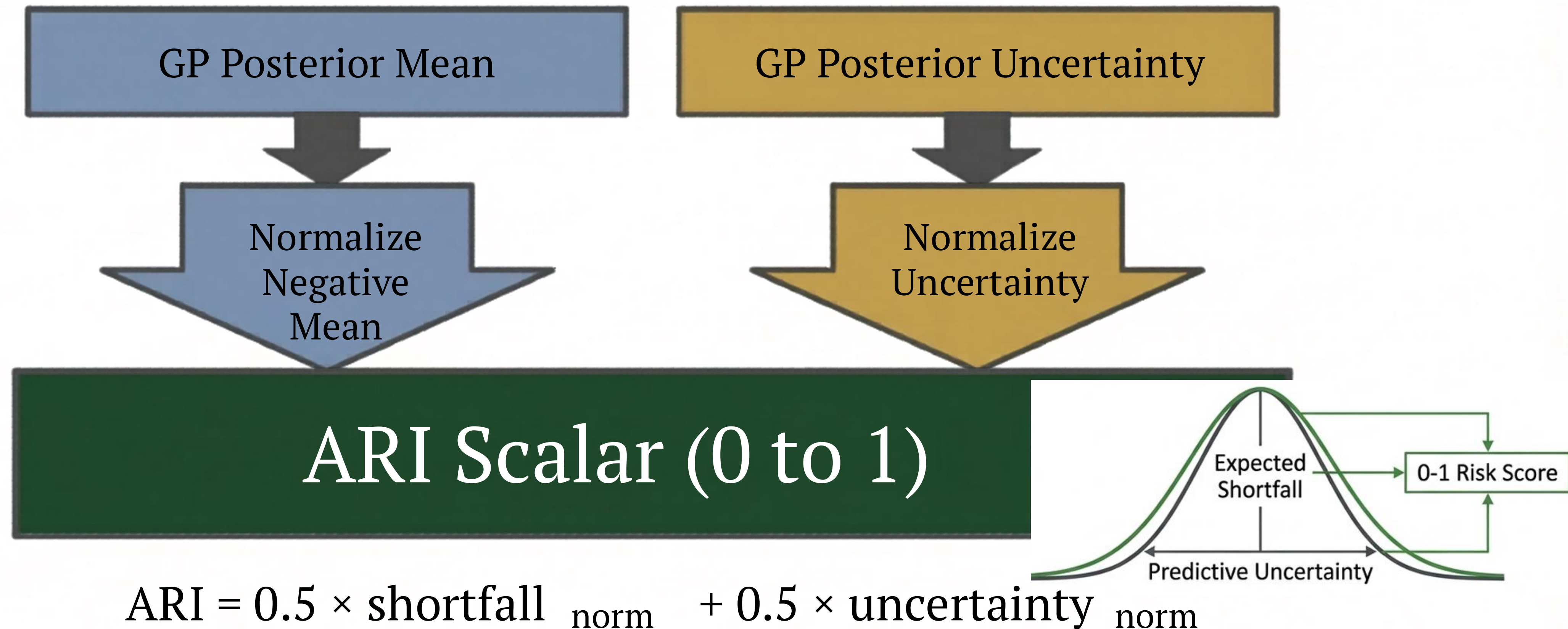


Year-Type Performance SPEI-12



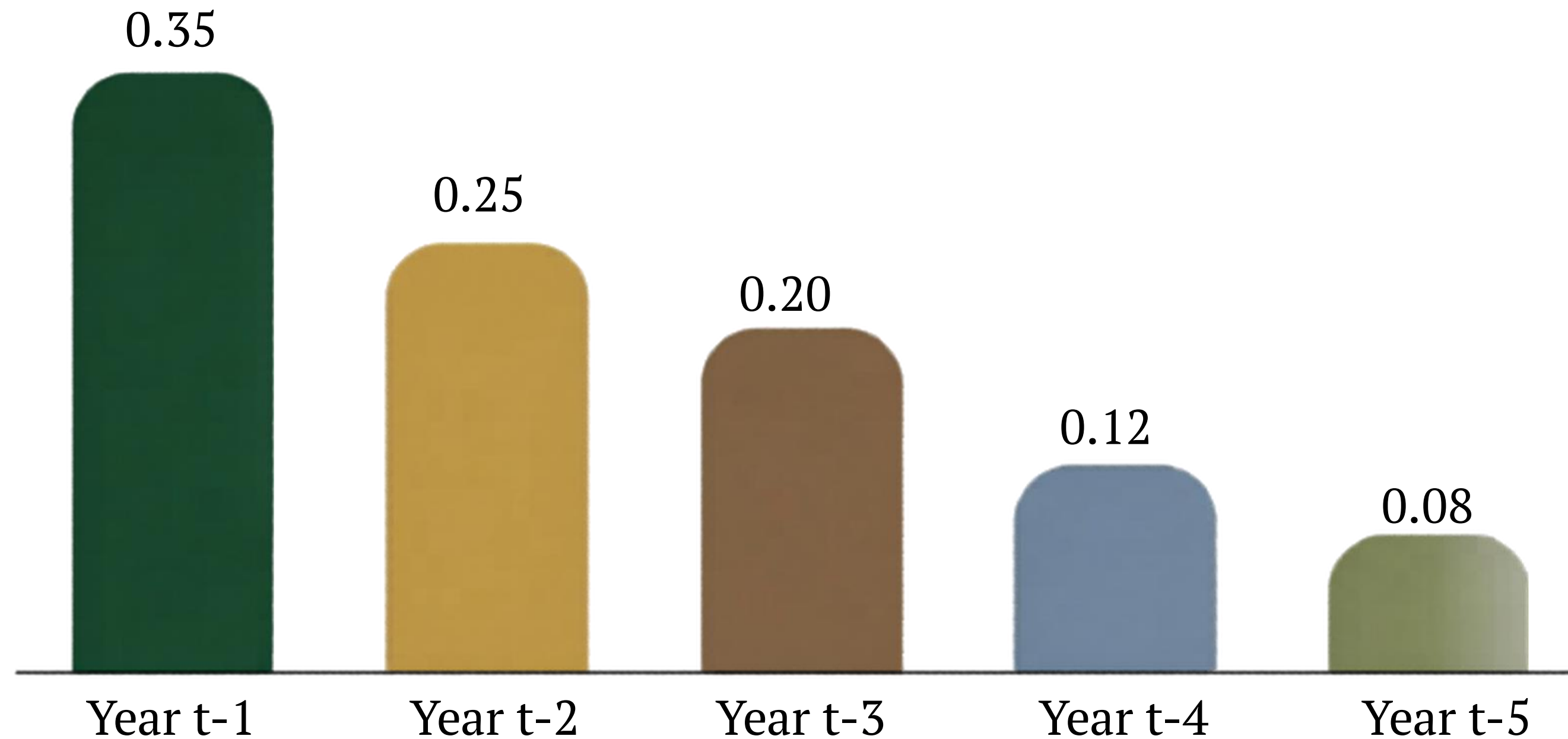
Actionable Extremes: Highest accuracy in dry years (SPEI-12 < -0.5) precisely when yield shortfalls drive the highest demand for conservation intervention

The Agricultural Risk Index (ARI)



- Equally weighting expected loss and predictive uncertainty
- A county with a guaranteed deficit and a county with massive unpredictability both need immediate policy attention

The Conservation Effectiveness Score (CES)



- Filters for yield-relevant NRCS practices, normalized by harvested acreage.

- Applies a 5-year decaying weight structure.

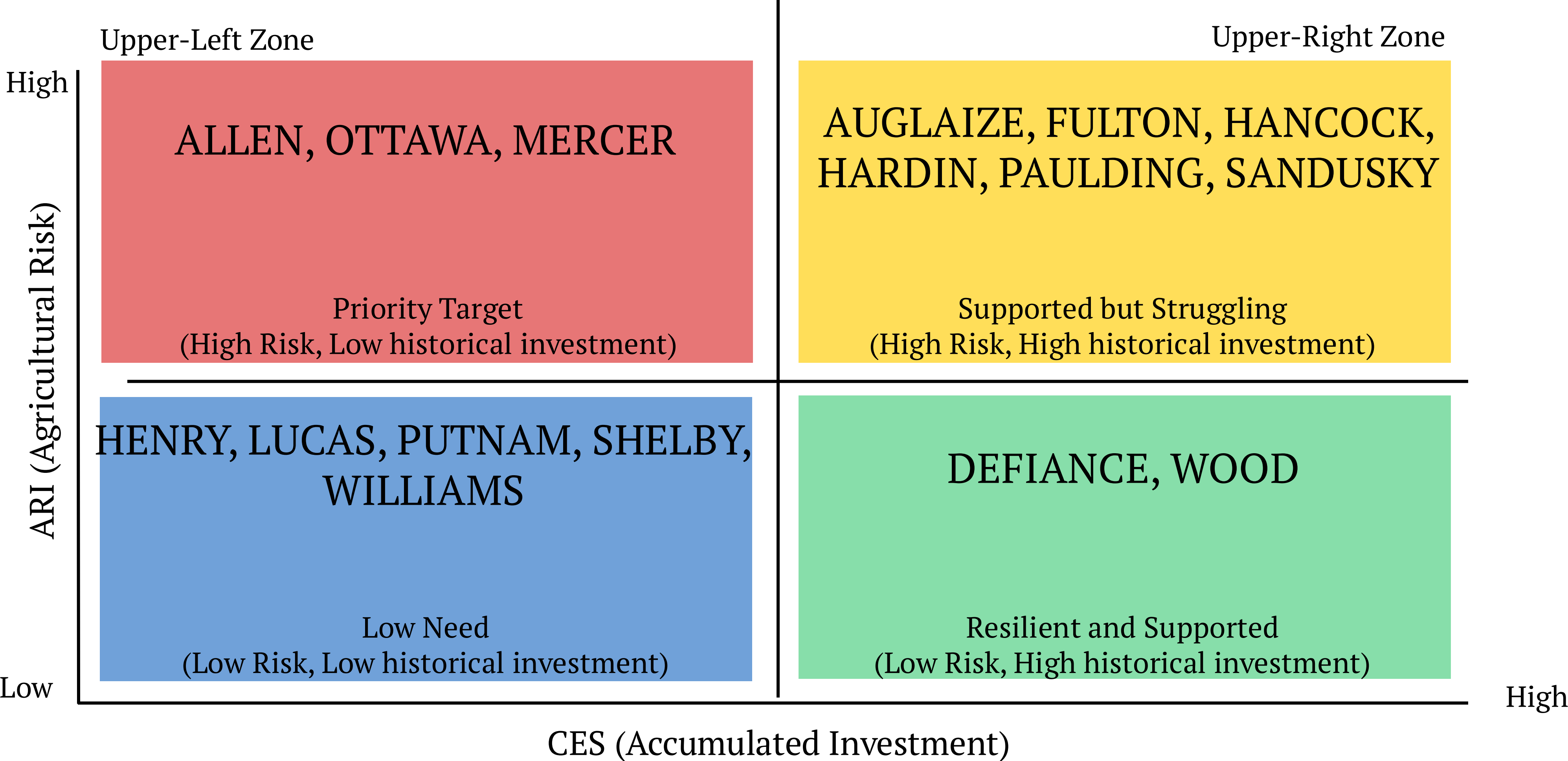
- Because soil health takes time to build, we don't just measure this year's money. We measure the accumulated, still-active conservation capital.

$$CES_t = \sum_{j=1}^5 w_j \cdot \frac{F_{t-j}}{A_{t-j}}$$

Outline

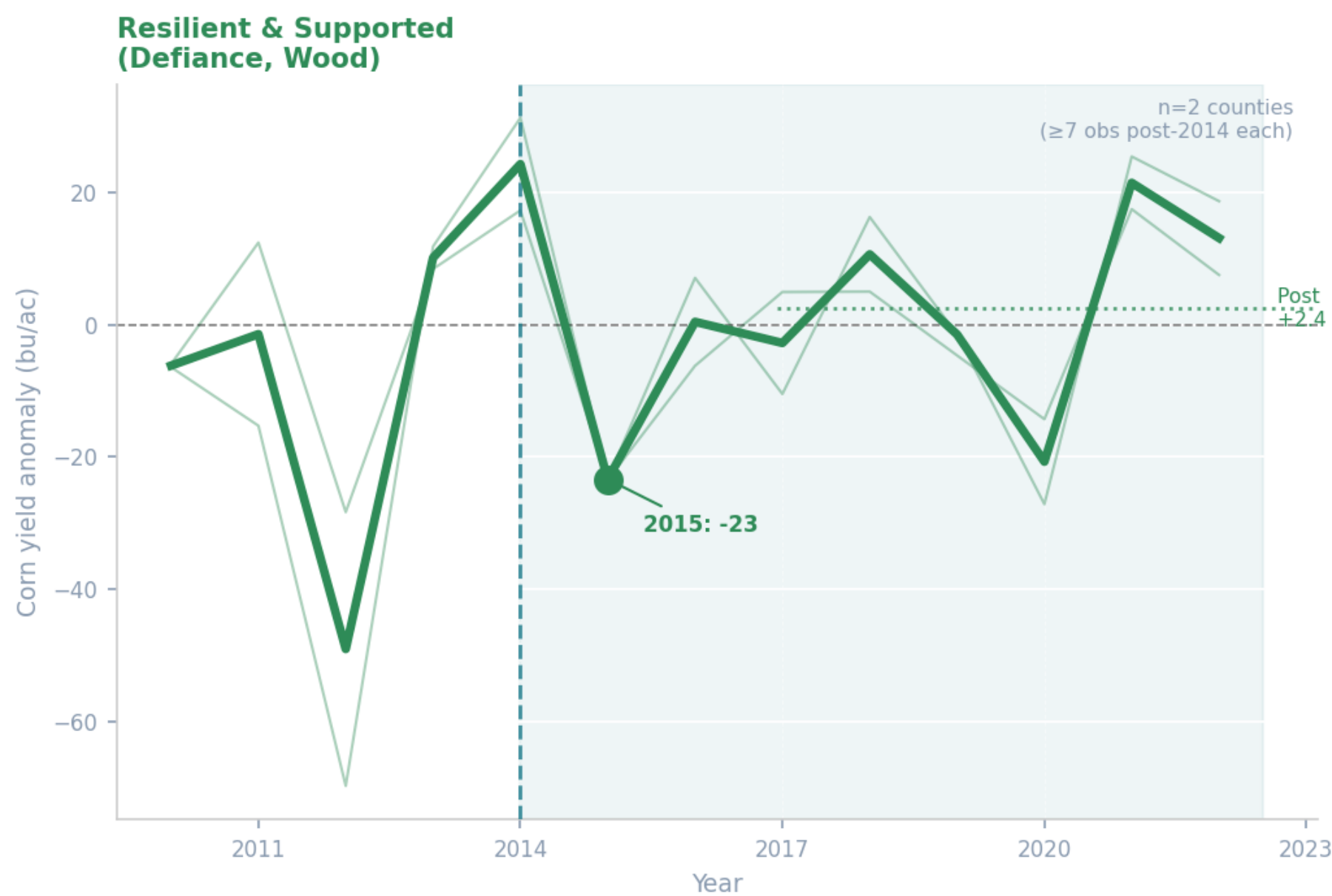
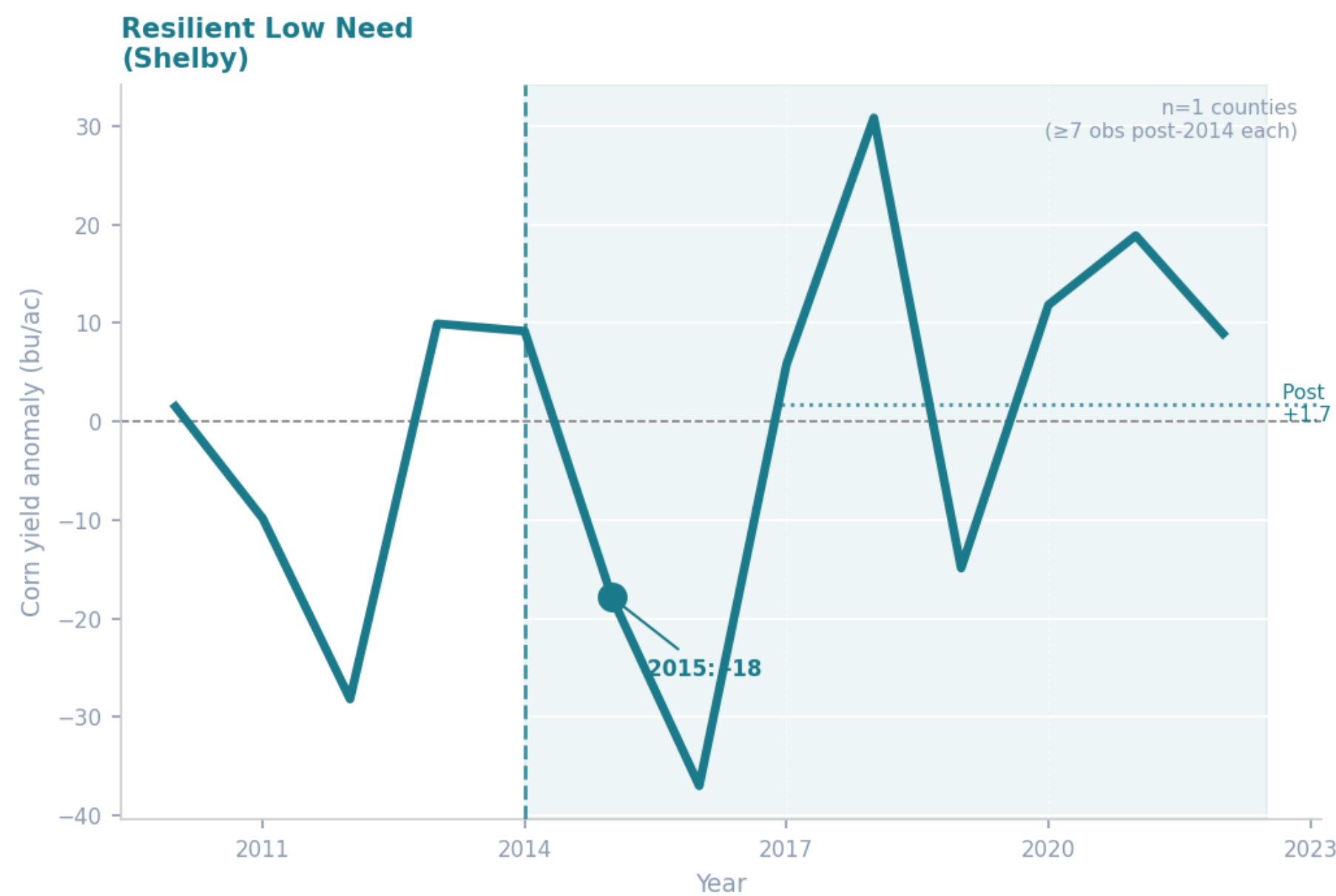
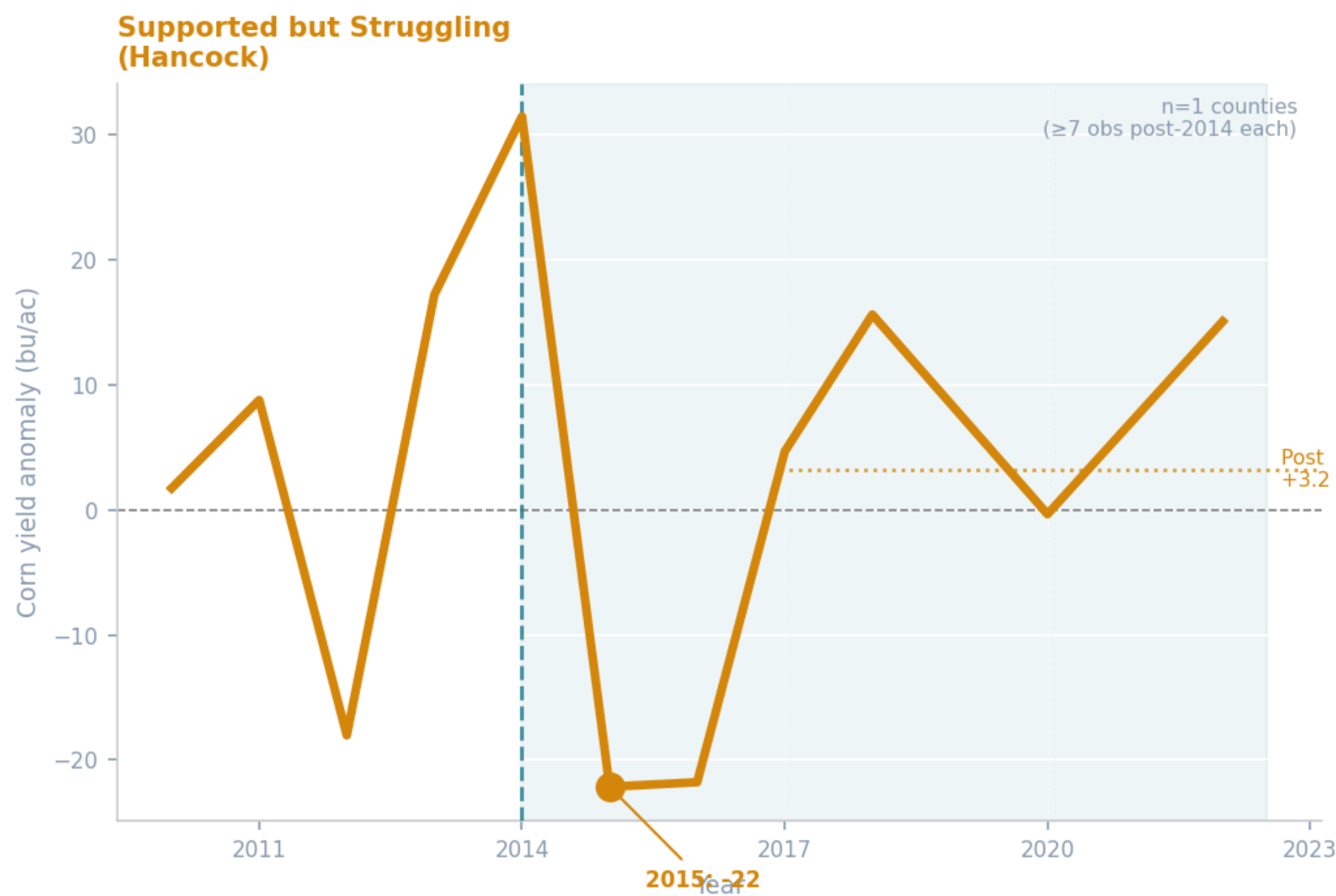
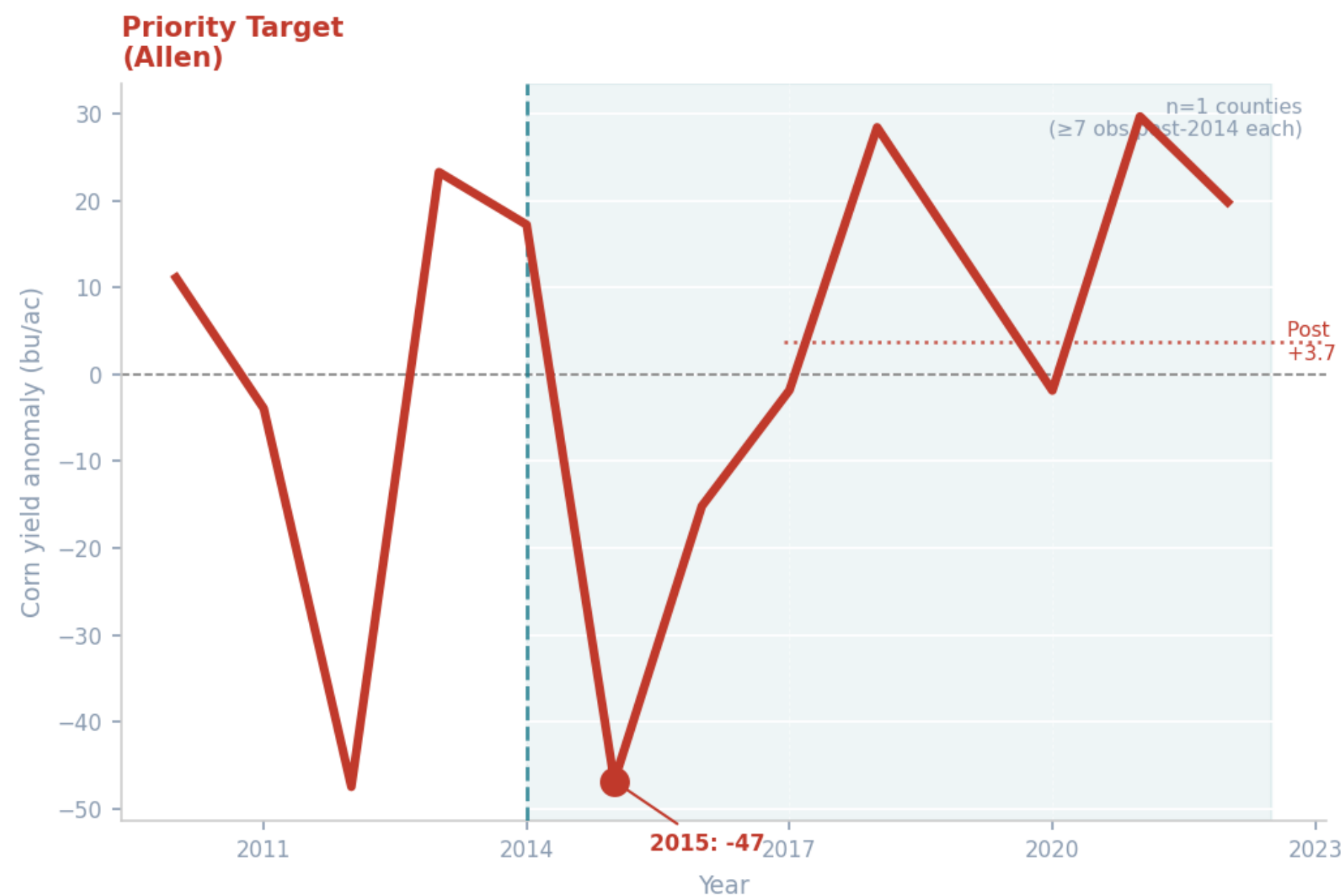
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The Conservation-Agriculture Gap Quadrant

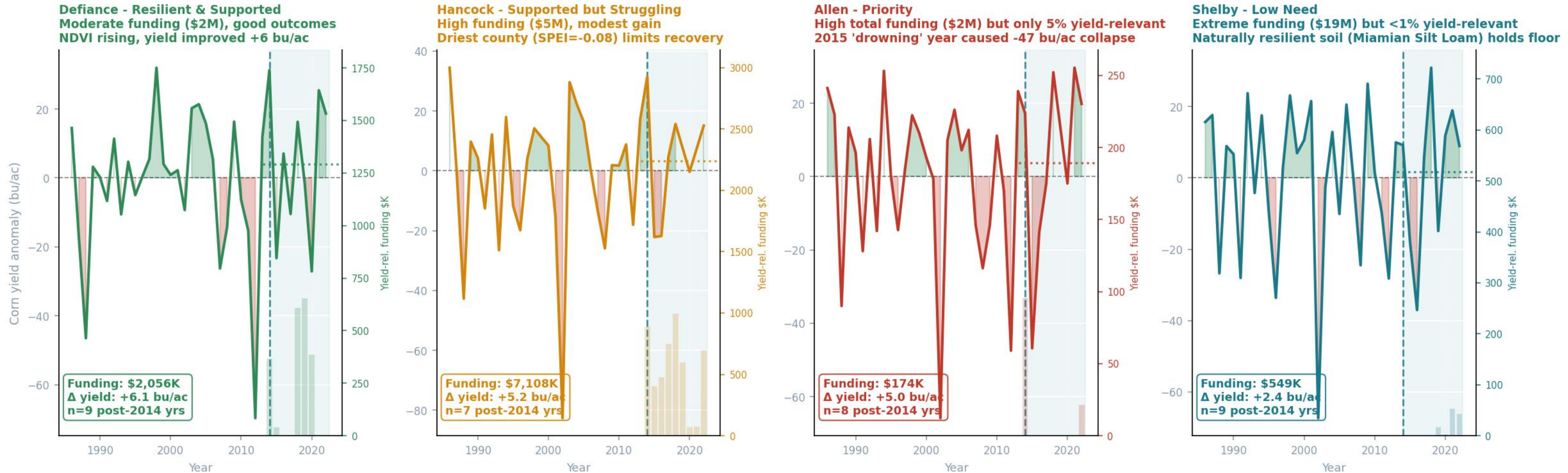


Transforming 36 years of mathematical prediction and isolated financial data into an actionable county typography

How Each Quadrant Group Responded to Stress Years (2015 & 2019)
Only counties with 7+ post-2014 observations; heavy line = group mean

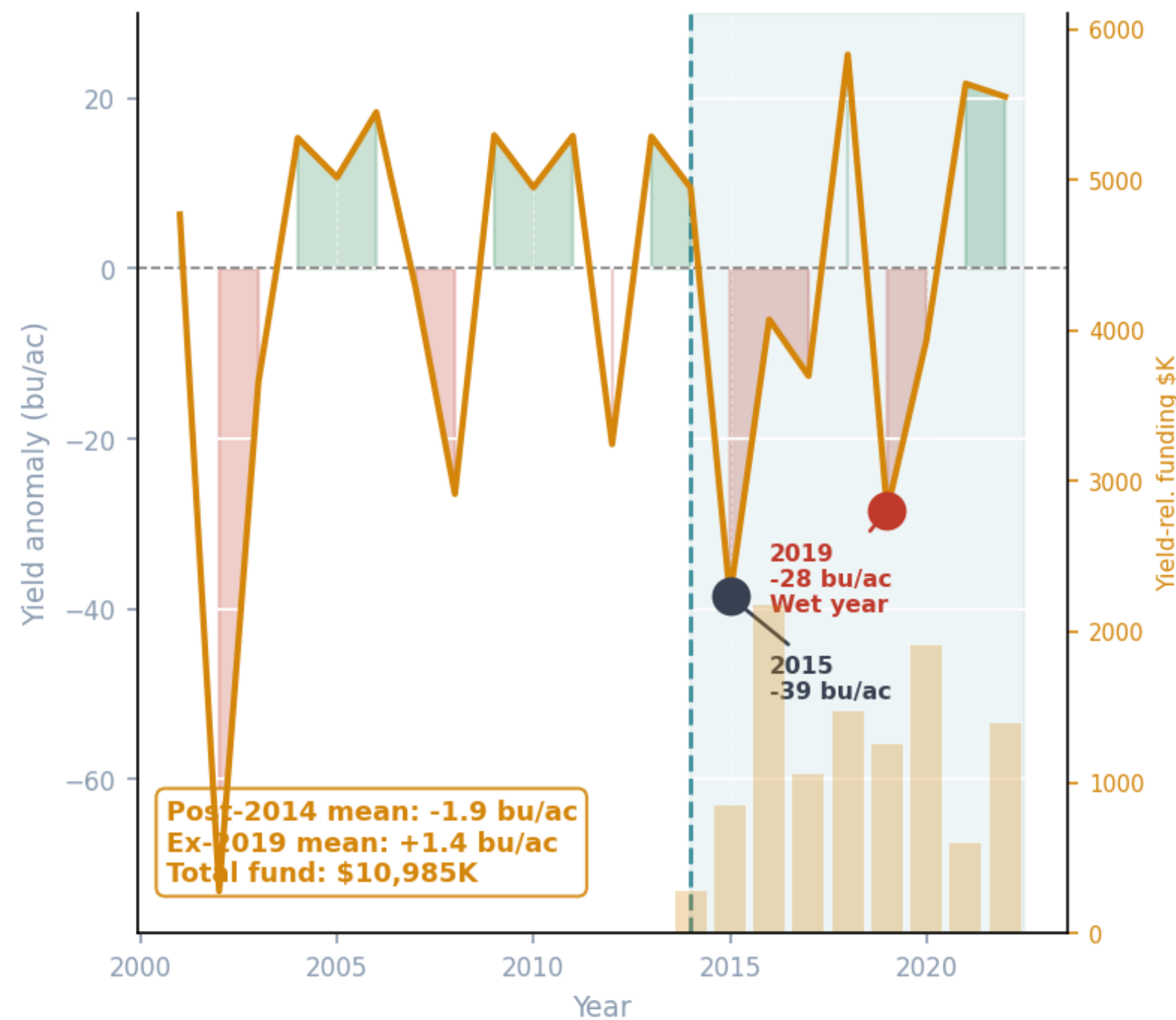


Same NRCS Era - Four Contrasting Outcomes

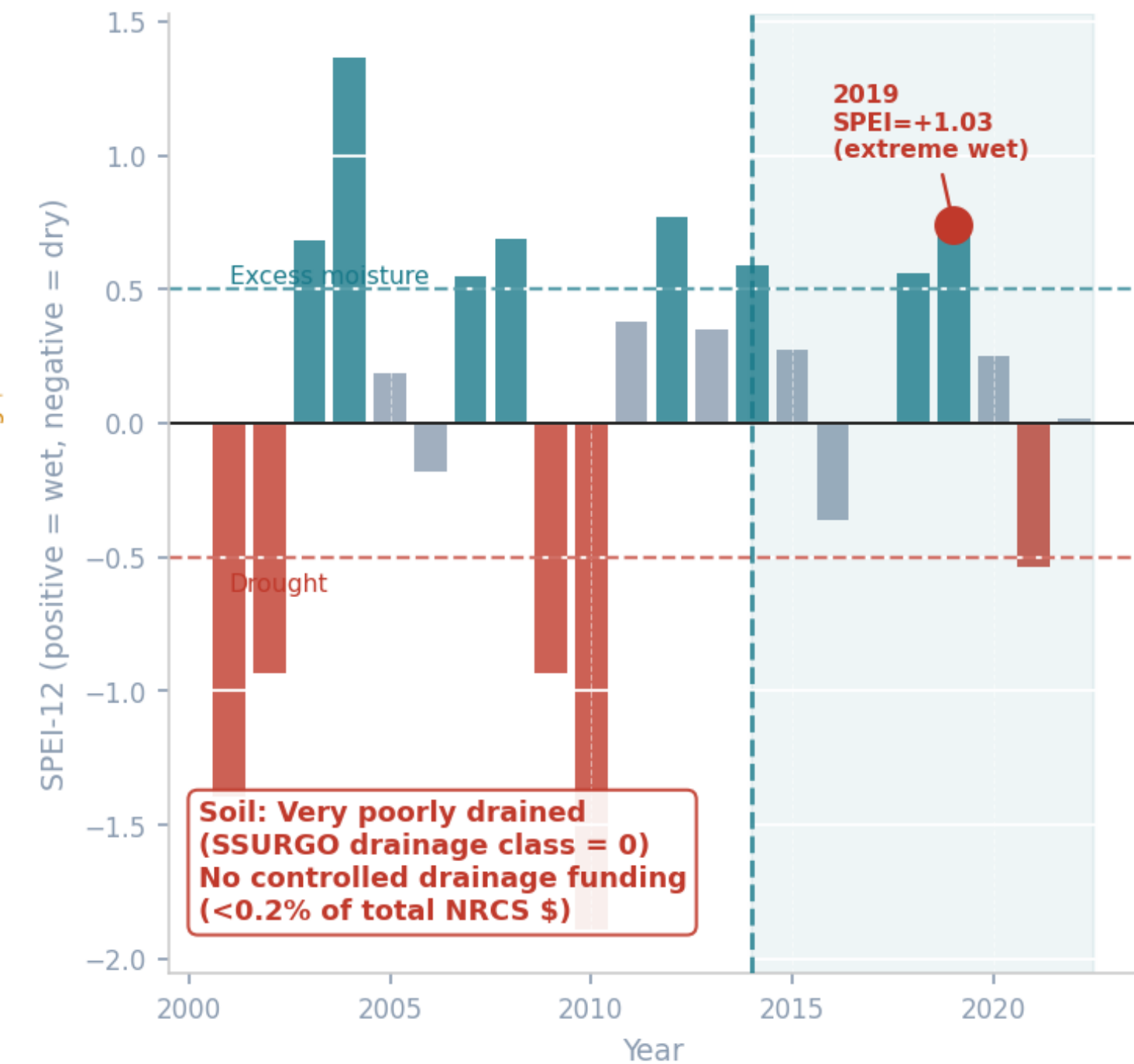


Hardin County (Supported but Struggling) - \$11M Invested, But 2019 Undid Years of Progress

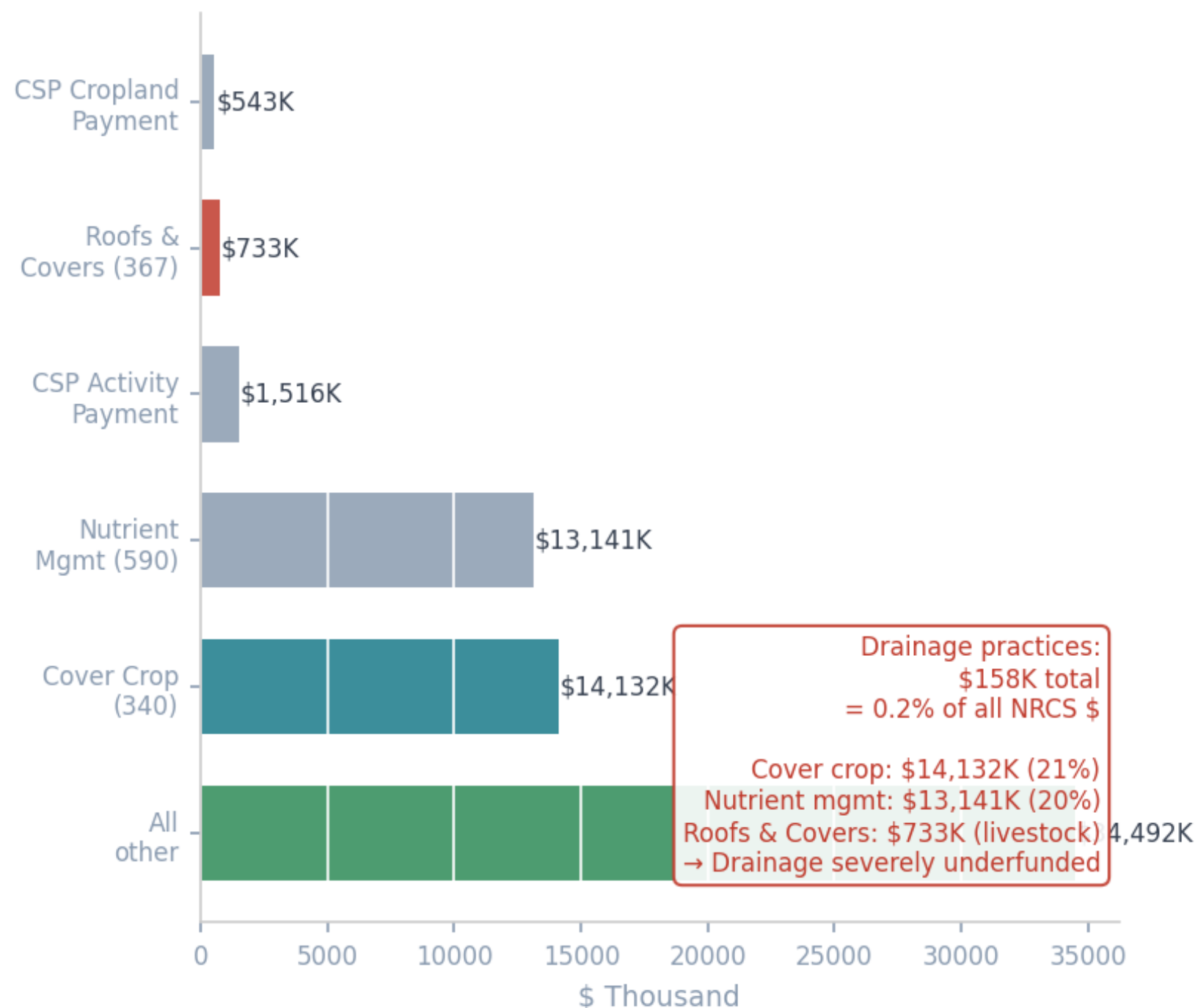
A - Yield Anomaly + Annual Funding
(One bad year = 2019)



B - Water Stress Index (SPEI-12)
(Hardin's risk = too much water, not too little)



C - Where Did Hardin's \$67M in NRCS Funding Go?
(Drainage = only 0.2% - yet drainage is the #1 risk)



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Multi-Disciplinary Thesis Contributions



Computer Science

- Domain-decomposed multi-kernel GP
- Reproducible, fusion-based public data pipeline
- Prediction-to-policy translation mechanism



Agriculture & Policy

- Yield forecast with calibrated uncertainty
- Delay-aware funding evaluation index (CES)
- Actionable 2x2 targeting typography

Who Can Use It?

- NRCS staff & TSPs
- Policymakers
- Extension staff, Researchers
- Grain experts

How They Can Use It?

- Annual county screening & targeted outreach
- Identify risk-funding gaps
- Farmers outreach & scaling model
- Anticipate production swings

Limitation

- County-level resolution
- Public data constraints
- Association-based policy insights
- Model Flexibility
- Scalability

Future Scope

- Finer spatial analysis
- Integrate farm-practice data
- Stronger causal validation
- Bayesian models
- Automated pipeline

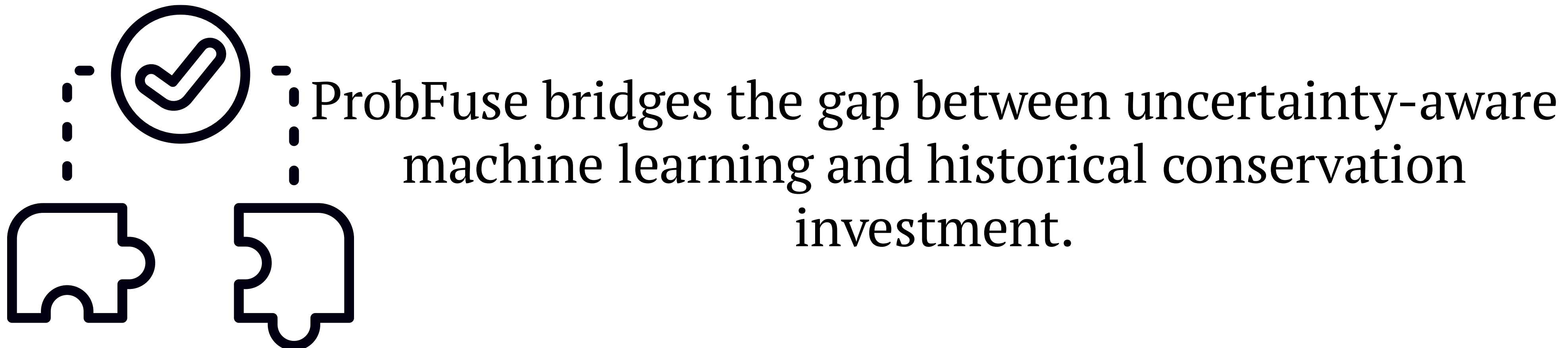
Conclusion

Is our conservation money
reaching the counties that need it
most?



Conclusion

Is our conservation money
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Thank you!